EE2021: Tutorial - 4 (Sept 18th, 2025)

1. Consider a system at 300 K which can exchange electrons through an ohmic contact. It consists of three single-particle states, A,B and C whose energy levels are such that $\epsilon_A > \epsilon_B > \epsilon_C$. The probability of A being occupied by a single electron is 0.047. The probability of B and C being empty are 0.5 and 0.047 respectively. Write the grand partition function (GPF) of the system (Quiz1 problem- Please go through Statistical mechanics slides)

2. Which of the following differential equations is linear?

(i) $z\frac{d\psi(z)}{dz} + g(z)\psi(z) = 0$, where $g(z)$ is some specific function

(ii) $\psi(z)\frac{d\psi(z)}{dz} + \psi(z) = 0$

(iii) $\frac{d^2\psi(z)}{dz^2} + b\frac{d\psi(z)}{dz} = c\psi(z)$, where b and c are constants

(iv) $\frac{d^3\psi(z)}{dz^3} = 1$

(v) $\frac{d^2\psi(z)}{dz^2} + \left(1 + |\psi(z)|^2\right)\frac{d\psi(z)}{dz} = g\psi(z)$, where g is a constant

3 Consider a one-dimensional particle which moves along the x-axis and whose Hamiltonian is $\hat{H} = -\mathcal{E}d^2/dx^2 + 16\mathcal{E}\hat{X}^2$, where $\mathcal{E}$ is a real constant having the dimensions of energy.

(a) Is $\psi(x) = Ae^{-2x^2}$, where A is a normalization constant that needs to be found, an eigenfunction of $\hat{H}$? If yes, find the energy eigenvalue.

(b) Calculate the probability of finding the particle anywhere along the negative x-axis.

(c) Find the energy eigenvalue corresponding to the wave function $\phi(x) = 2x\psi(x)$.

4 A particle of mass m, which moves freely inside an infinite potential well of length a, has the following initial wave function at $t = 0$:

$$\psi(x, 0) = \frac{A}{\sqrt{a}}\sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{3}{5a}}\sin\left(\frac{3\pi x}{a}\right) + \frac{1}{\sqrt{5a}}\sin\left(\frac{5\pi x}{a}\right),$$

where A is a real constant.

(a) Find A so that $\psi(x, 0)$ is normalized.

(b) If measurements of the energy are carried out, what are the values that will be found and what are the corresponding probabilities? Calculate the average energy.

(c) Find the wave function $\psi(x, t)$ at any later time t.

(d) Determine the probability of finding the system at a time t in the state $\varphi(x, t) = \sqrt{2/a}\sin(5\pi x/a)\exp(-iE_5t/\hbar)$; then determine the probability of finding it in the state $\chi(x, t) = \sqrt{2/a}\sin(2\pi x/a)\exp(-iE_2t/\hbar)$.

5. Consider an infinite potential well of width 2a with a Dirac delta function ($\boldsymbol{\alpha}\delta(x)$) in the middle of the well.

a) Write the Schrodinger equation, apply boundary conditions and derive the expression for wavefunction and the bound states for $\boldsymbol{\alpha} > 0$

b) Do the same for $\boldsymbol{\alpha} < 0$ (polarity of delta function well is flipped)

c) Without solving the Schrodinger equation, try to sketch the lowest two eigenfunctions if the Dirac delta function is replaced by an energy barrier of finite height and width (2b, where b <a)

